Forces in 2D & Vector Notation

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Forces in 2D & vector notation

How are forces and vectors related?

- **Forces** are **vectors** – they have **magnitude** and **direction**.
- The **Magnitude** of a force is measured in Newtons and the **direction** of a force is an **angle**, usually measured in **degrees**, **anti-clockwise** from the **horizontal**
- There are other ways to talk about direction – in particular **Cartesian coordinates**(x, y) are used to describe the position of a point in two-dimensional space (plane), relative to a fixed **origin O.**
- You may see questions start with a sentence along the lines of "Relative to the frame of reference **Oxy** …" – this means the two-dimensional space (plane) and distances within it will be based on **horizontal** (x) and **vertical** (y) components relative to the origin O.

What is vector notation?

- **Both column vectors** and **i, j notation** can be used for calculating **resultant vectors**
- In both cases, a vector (force) is written with its two components separated so the **magnitude** and **direction** of the vector are **not** directly known.

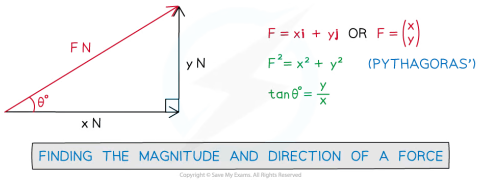
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

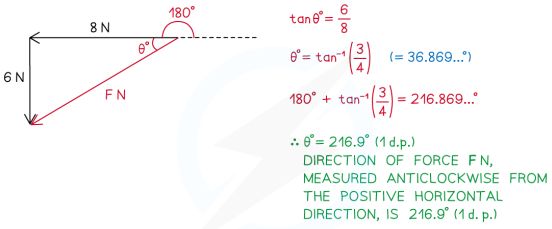
- The **magnitude** of the force **F N** would be denoted by |**F**| N or *F N* – notice the use of bold and italics in particular
- **Direction** is an angle, usually measured in degrees anti-clockwise from the horizontal, *θ°* is usually used

How do I find the magnitude and direction of a force from its components?

- For a force **F** = x**i** + y**j** N to find
  - its **magnitude**,|**F**| N or F N ,use Pythagoras' theorem
  - its **direction** (as an angle), use a diagram and trigonometry



- If either (or both) components are **negative** then still use a diagram and trigonometry but
  - treat *x* and *y* as **positive** (so |x| and |y| strictly speaking)
  - the angle found may need **adjusting** depending on where, and which way, the **direction** is being **measured** from
  - e.g. Find the direction at which the force **F** = (-8**i** - 6**j**) N, giving your answer as an angle measured in degrees anti-clockwise from the positive horizontal direction



What does equilibrium mean with vectors?

- In two dimensions a **particle** is in **equilibrium** if the **resultant force** acting on it in both directions is **zero**
  - For **column vectors**, a **resultant force** of **zero** would look like $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$
  - For vectors in **i–j notation**, a **resultant force** of zero would look like (0i + 0j) N

Worked Example

Two forces, $(3\mathbf{i} + 8\mathbf{j})$ N and $(-5\mathbf{i} - 11\mathbf{j})$ N act on a particle.

(i) Find the resultant force.

(ii) Find the magnitude of the resultant force and its direction as an angle measured anti-clockwise from the **i**-direction.

(iii) A third force is applied to the particle such that it is brought into equilibrium. Find the third force, giving your answer in the form $(x\mathbf{i} + y\mathbf{j})$ N

Answer:

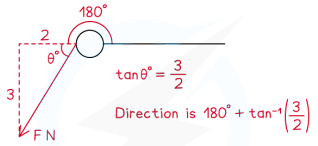
(i) Let the resultant force be $\underline{F}$ N.

$\underline{F} = (3\underline{i} + 8\underline{j}) + (-5\underline{i} - 11\underline{j})$

$\therefore \underline{F}\text{ N} = (-2\underline{i} - 3\underline{j})\text{ N}$

(ii) $F^2 = (-2)^2 + (-3)^2 = 13$

Draw a diagram to help find the required direction



$\tan\theta° = \frac{3}{2}$

Direction is $180° + \tan^{-1}\left(\frac{3}{2}\right)$

$\therefore$ Magnitude of $\underline{F}$ N is $F\text{ N} = \sqrt{13}$ N

The direction of $\underline{F}$ N is 236.3° (1 d.p.) anti-clockwise from the $\underline{i}$-direction

(iii) From part (a) we know the first two forces give a resultent force $\underline{F} = (-2\underline{i} - 3\underline{j})$N

$\therefore \underline{F} + (x\underline{i} + y\underline{j}) = 0$ $\quad (\underline{0} = 0\underline{i} + 0\underline{j}$, the 'zero vector')

$(-2 + x) = 0, \quad x = 2$

$(-3 + y) = 0, \quad y = 3$

$\therefore$ The third force is $(2\underline{i} + 3\underline{j})$N

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Examiner Tips and Tricks

- Ideally you should stick to the **i-**,**j-** vector notation used in a question but if you prefer one over the other you can use a mixture within the same question. Just be careful your final answer is in the correct format if requested.
- Draw diagrams – including 'mini'-diagrams of individual forces/vectors – this can help in understanding a problem and being accurate.

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